

De-spike and Interface-Plane Closure: A Mechanistic Decomposition of SolarLab's Heterojunction Trend Response vs SCAPS-1D

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Abstract

This note records the outcome of a controlled four-configuration experiment that isolates *which* heterojunction mechanism each parameter-sweep trend depends on. Two independent corrections were tested — the **heterointerface Auger de-spike** (het_recomb_despike, a bulk-rate blend that removes the over-counted Auger recombination at the junction mesh node) and the **interface-plane closure** (interface_plane_closure, a per-RHS quasi-steady algebraic interface-state term) — both alone and combined, against a steady-state interface-states reference. The result is a clean separation: **the de-spike governs the base V_{oc} and the bulk-defect (N_t) trends; the closure governs the interface-defect directions; the two are orthogonal and complementary, not redundant**. Combining them (**configuration AB**) is the best fast-transient parity point measured — base $V_{oc} = 1.161$ V (within 7 mV of SCAPS) with all bulk and interface trends in the correct direction. Two residuals remain and are individually named and bounded: a $4\times$ over-response of the near-passive HTL/PVK interface N_t trend (a de-spike/closure overlap at one shared node), and the long-standing ETL-doping (N_d) direction mismatch (resolved only by the steady-state interface states, which are presently $\sim 80\times$ too slow to use in the sweep harness).

1. The four configurations

All share the reference Base Model (configs/scaps_mirror_v2.yaml, dos_band_potentials: true). They differ only in which interface correction is active.

ID	het_recomb_despike	interface_plane_closure	Driver	Cost
A	0.53	off	transient run_jv_sweep	~7 s/point
B	0	on	transient run_jv_sweep	~7 s/point
AB	0.53	on	transient run_jv_sweep	~7 s/point
C	0	off (steady-state iface_states)	Newton solve_voc_ss	~480 s/ V_{oc}

Table 1. The de-spike fraction $f = 0.53$ is the base-optimal value from the prior de-spike matrix sweep (it lands base V_{oc} on SCAPS; f monotonically lifts V_{oc} from 1.118 V at $f = 0$ to 1.201 V at $f = 1$). Configuration C is the steady-state interface-states solver — included as the physical reference for the interface directions, but impractical in the sweep harness at its current cost (a single V_{oc} took 481 s; the full 85-point matrix would be ~11 h).

2. Base operating point

Config	V_{oc} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)	ΔV_{oc} vs SCAPS
A (de-spike only)	1.1638	25.72	87.32	26.14	-3.8 mV
B (closure only)	1.1178	25.70	88.23	25.35	-49.8 mV
AB (both)	1.1610	25.72	88.54	26.44	-6.6 mV
C (SS states)	1.1063	—	—	—	-61.3 mV
SCAPS-1D	1.1676	26.28	86.99	26.69	—

Table 2. The de-spike is what closes the base V_{oc} : removing the over-counted junction Auger lifts B's 1.118 V to A's 1.164 V and AB's 1.161 V. The closure alone (B) does almost nothing to the base point; the steady-state states (C) actually sit *below* B at the base, confirming that the base deficit is a bulk-recombination over-count, not an interface-transport effect.

3. Trend matrix

Closure % is the dimensionless ratio (SolarLab V_{oc} range / SCAPS V_{oc} range) over each sweep; OK = correct direction, X = direction mismatch, “flat/flat” = both solvers respond < 1 mV so the ratio is meaningless-but-matched.

Sweep	SCAPS ΔV_{oc} (mV)	A (de-spike)	B (closure)	AB (both)
CBO (χ_{ETL})	918	85 % OK	77 % OK	83 % OK
N_d (ETL doping)	100	25 % Xdir	11 % Xdir	28 % Xdir
Nt PVK (CB)	38.6	69 % OK	11 % OK	63 % OK
Nt PVK (VB)	10.8	248 % OK	41 % OK	227 % OK
Nt HTL/PVK	5.2	0 % Xflat	70 % OK	425 % OK (4× over)
Nt PVK/ETL	282	72 % OK	45 % OK	64 % OK
Et PVK (CB)	0.4	flat/flat	flat/flat	flat/flat
Et PVK (VB)	0.4	flat/flat	flat/flat	flat/flat
Et HTL/PVK	0.006	flat/flat	flat/flat	flat/flat
Et PVK/ETL	34.5	23 % OK	6 % OK	32 % OK

Table 3. Reading the columns: **A** (de-spike) carries the bulk N_t trends (69 %, 72 %) but cannot move the near-passive HTL/PVK interface N_t at all (flat, wrong direction). **B** (closure) is the only column that gives the HTL/PVK interface N_t trend physically (70 %, matching SCAPS's small 5.2 mV range) but leaves every bulk N_t trend weak (11 %, 41 %). **AB** keeps both — every bulk and interface trend points the right way — at the price of one over-response (next section).

4. The two-bottleneck model

The matrix decomposes cleanly into two independent suppression mechanisms.

Bottleneck 1 — over-counted junction Auger (governs N_t response and base V_{oc}). On a continuous mesh, the heterojunction node sees a $\sim 1000\times$ carrier pile-up from the band offset (Section 6 of the gap analysis). The Auger rate $\propto n^2p, np^2$ then explodes at that single node and dominates total recombination, so any *additional* bulk defect N_t the user adds is masked — its SRH contribution is a rounding error next to the artifact Auger. The de-spike blends the junction-node carrier densities toward the neighbour geometric mean *in the bulk recombination rate only* (transport untouched), removing the artifact. The effect is visible directly in Table 3: without de-spike (B) the bulk N_t trends are 11 % / 41 %; with it (A, AB) they jump to 69 % / 248 %. **The artifact junction Auger is the single dominant bottleneck for the N_t response.**

Bottleneck 2 — bulk-node sampling of E_t (governs E_t response). The SRH rate

$$R_{SRH} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \quad n_1 p_1 = n_i^2 e^0, \quad n_1 = n_i e^{(E_t - E_i)/kT}$$

carries the trap energy E_t only through n_1, p_1 . At a junction node where n, p are huge from the pile-up, $n \gg n_1$ and $p \gg p_1$ for any E_t , so the E_t -bearing terms vanish from the denominator and the rate becomes E_t -independent. De-spike the carrier densities partially restores sensitivity (Et PVK/ETL: B = 6 % $\rightarrow$ A = 23 % $\rightarrow$ AB = 32 %), **but it caps around 32 %** because the de-spike node is still a single bulk node, not a true interface state with its own occupancy. Closing the remaining E_t gap requires the interface-plane states (the steady-state path, configuration C / the transient closure extended). **So E_t has two bottlenecks — the Auger over-count and bulk-node sampling — and the de-spike only removes the first.**

5. Residual 1 — HTL/PVK interface N_t over-response

This is the one place AB is worse than its parts. The HTL/PVK interface is near-passive: SCAPS moves only 5.2 mV across the whole N_t sweep. Per Table 3:

- **A (de-spike only):** 0.0 mV — completely flat, wrong direction. The de-spike has no interface-state term, so it cannot respond to an interface defect.
- **B (closure only):** 3.64 mV — 70 % closure, a *good* match to SCAPS's 5.2 mV. The closure term alone reproduces this interface correctly.
- **AB (both):** 22.15 mV — 425 % closure, **4 $\times$ the SCAPS range.**

The mechanism: the closure exposes the interface- N_t sensitivity at the junction node, and the de-spike, by stripping the bulk-Auger floor at that *same shared node*, removes the very thing that was damping it — so on this nearly passive interface the combined response is amplified $\sim 6\times$ over closure-alone. This is a de-spike/closure **overlap at one mesh node**, not a physics error in either term separately. The fix is an interface-node-aware de-spike: skip the de-spike blend on nodes the closure already owns, so each shared node is corrected once. Expected to restore AB's HTL/PVK N_t to B's 3.6 mV match while leaving every other entry in the AB column unchanged.

6. Residual 2 — ETL doping (N_d) direction mismatch

Every transient configuration (A, B, AB) gets this backwards: as ETL donor doping rises, SCAPS V_{oc} climbs +100 mV monotonically, while SolarLab *falls* ~ 10 mV. The de-spike and closure both shift the magnitude (11 $\rightarrow$ 25 $\rightarrow$ 28 % closure) but none flip the sign. Only **configuration C (steady-state interface states)** reverses it (established in the prior root-cause work; root = the frozen built-in-potential boundary, which the steady-state interface treatment relaxes). C is physically correct here but presently $\sim 80\times$ too slow for the sweep harness (481 s per V_{oc}). This residual is therefore gated on a performance task, not a physics task: profile and accelerate the steady-state interface-states Newton solve until it can join AB.

7. Recommendation

1. **Adopt AB as the validated fast-transient parity config** — de-spike $f = 0.53 + \text{interface_plane_closure}$ on. It holds base V_{oc} 1.161 V (-7 mV), FF/PCE within tolerance, and every bulk and interface trend in the correct direction, at ~ 7 s/point.
2. **Implement the interface-node-aware de-spike** (Section 5) to retire the HTL/PVK N_t $4\times$ over-response — the only blemish in the AB column.
3. **Accelerate the steady-state interface states** (Section 6) so the ETL-doping direction can be fixed without the 11 h sweep cost.
4. **Treat E_t closure as state-limited** (Section 4): the de-spike caps E_t response at $\sim 32\%$; do not chase the rest with more de-spike — it needs the interface-plane states.

Appendix — reproduction

Artifact	Path
A/B/AB/C comparison driver	/tmp/abc_compare.py (--variant {A,B,AB,C})
Per-variant trend JSON	/tmp/abc_out/variant_{B,AB,C}.json
A = de-spike $f = 0.53$ full matrix	/tmp/despikes_out/partB_f0.53.json
De-spike f -sweep (base + trends)	outputs/despikes_matrix/despikes_matrix.csv
De-spike figures	outputs/despikes_matrix/fig_*.png
De-spike implementation	perovskite_sim/physics/continuity.py (het_recomb_despikes)
Closure / states flags	perovskite_sim/models/device.py; perovskite_sim/experiments/steady_state.py
Trend metric (closure %, dir, flat)	scripts/run_scaps_full_regression.py (trend_stats, verdict)